

QUANTUM COMPUTING

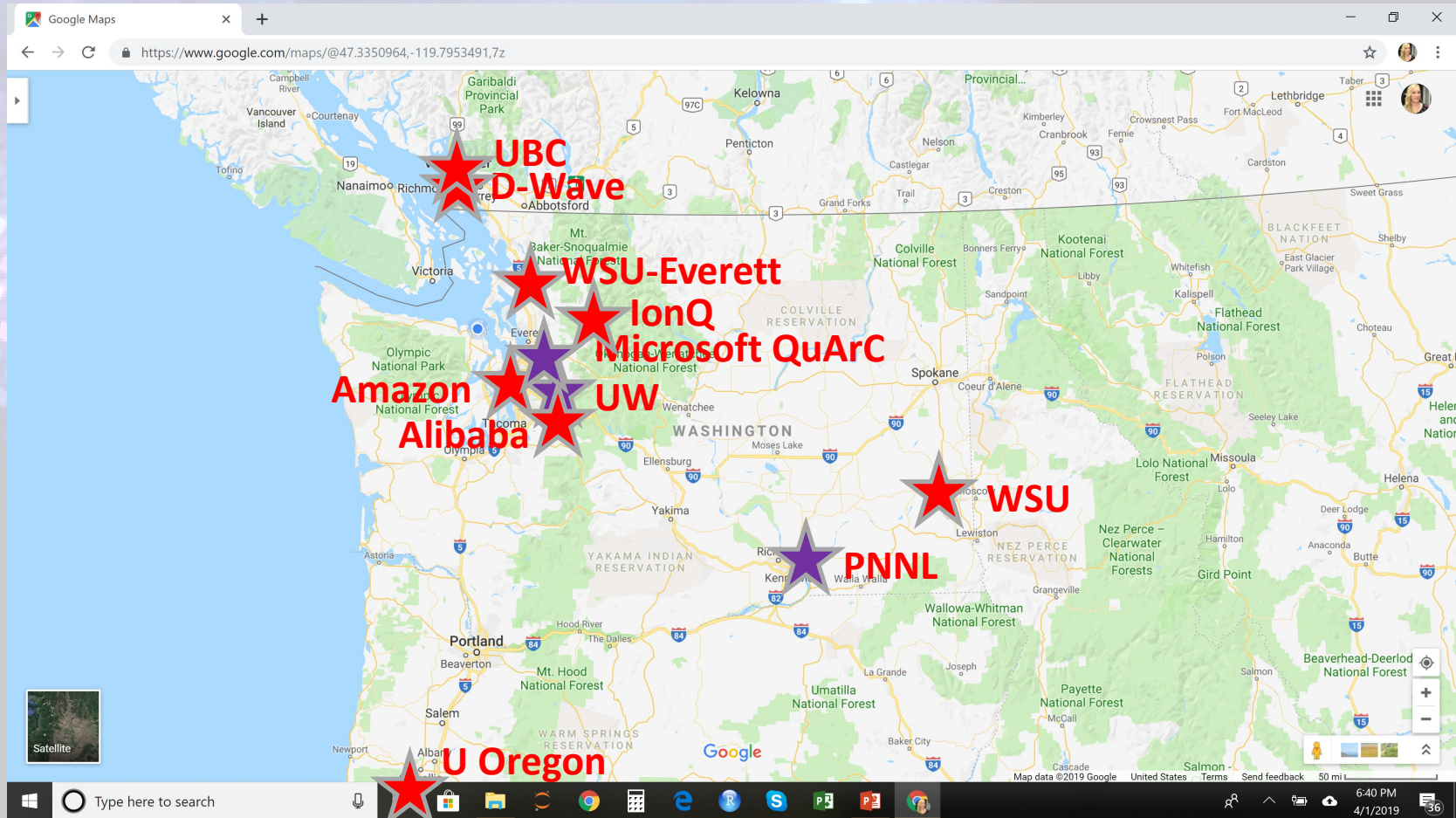
ITS TIME IS NOW (WK2-PT4)

Professor Nella Ludlow, PhD

Wright State University

Dayton, OH

NW QUANTUM NEXUS



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Congressman Derek Kilmer and Tom Alberg, Managing Director of Madrona Ventures.

Tom Alberg authored *Quantum Computing is Coming, Let's Focus on Getting our Computer Science Workforce Ready*



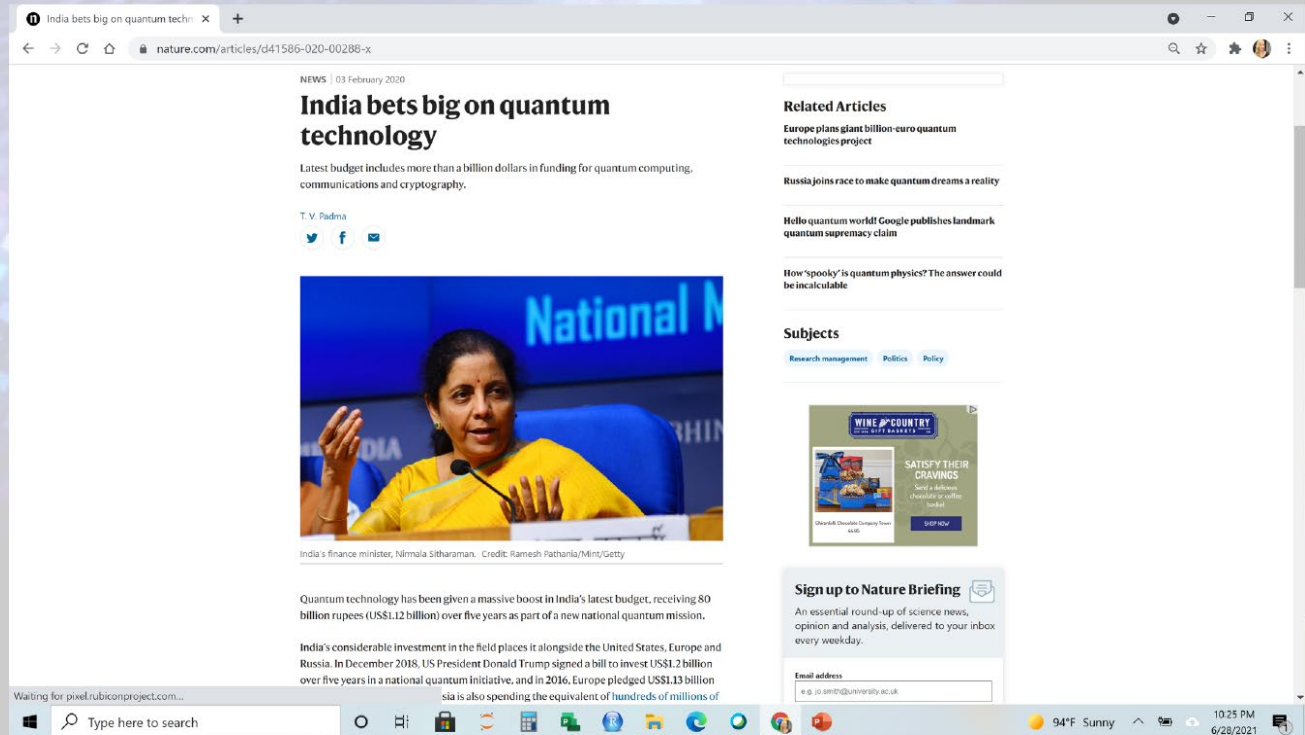
2012 Nobel physics laureate David Wineland, a professor at the University of Oregon, speaks during the Northwest Quantum Nexus summit on March 18, 2019.



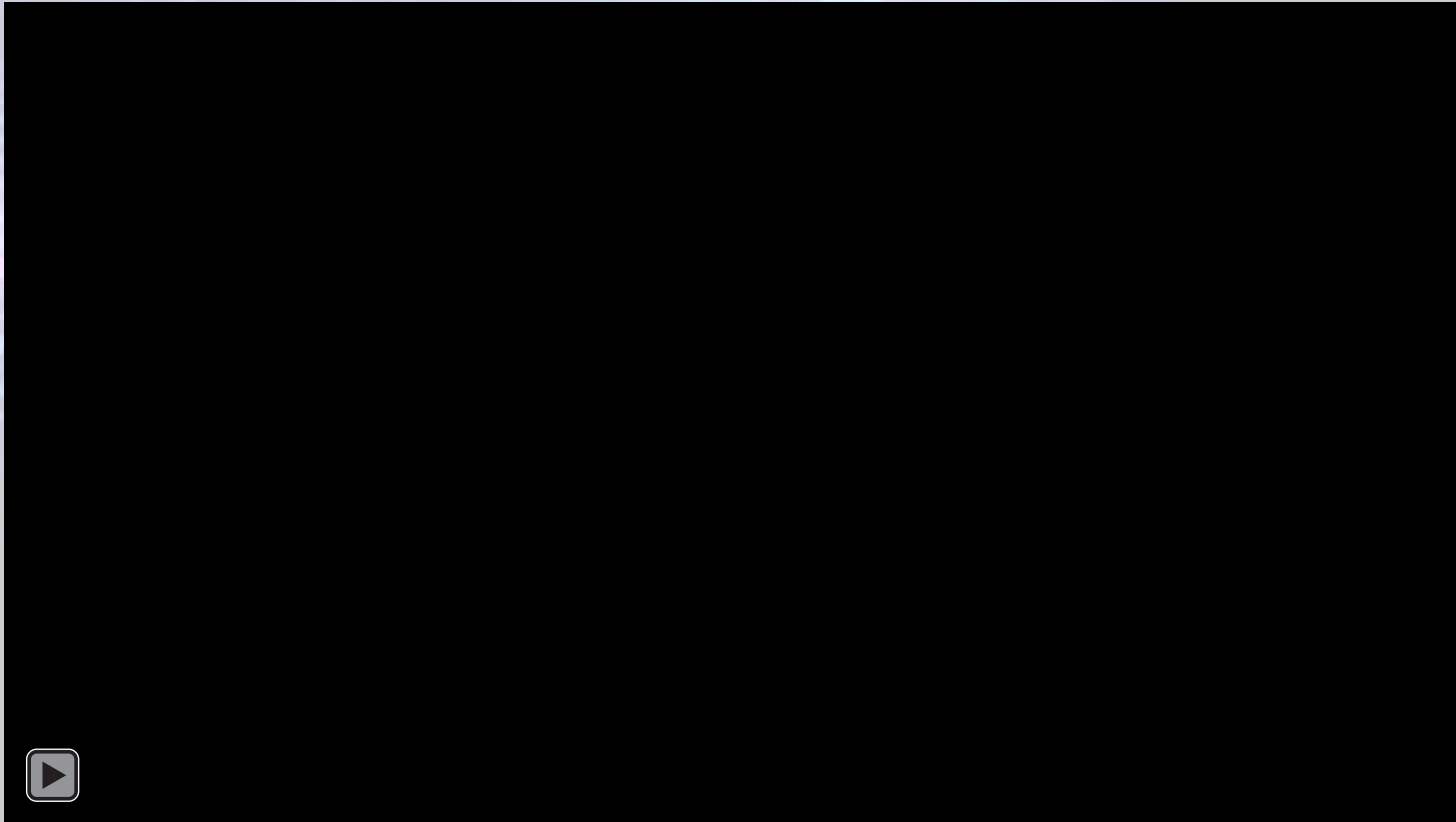
- The President signed into law the National Quantum Initiative Act (NQIA) December 21, 2018.
- Bill was approved by unanimous consent in the Senate and by a vote of 348-11 in the House.
- \$1.2 billion over five years
 - Investment in quantum information science
 - Development of a quantum-smart workforce
 - Establishes a National Quantum Coordination Office
 - Creates an advisory committee to advise the White House on Quantum Computing.

INDIA

- 80 Billion Rupees
(\$1.12B US)



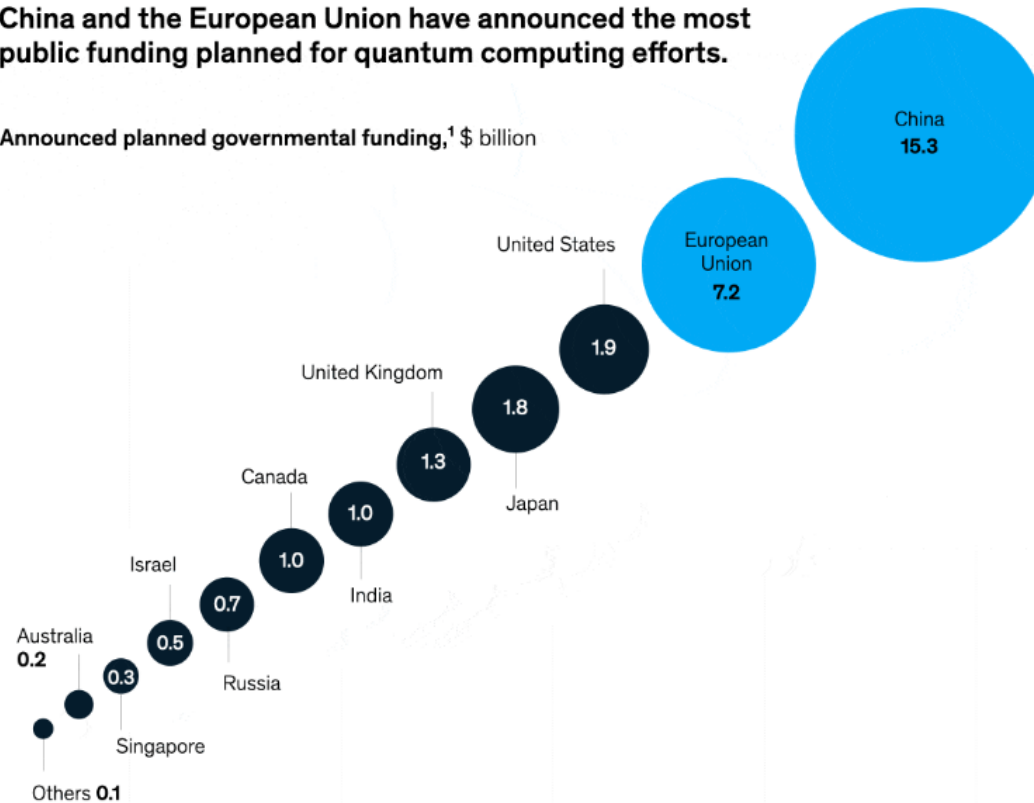
President Signs CHIPS and Science Act



Funding by Country

China and the European Union have announced the most public funding planned for quantum computing efforts.

Announced planned governmental funding,¹ \$ billion



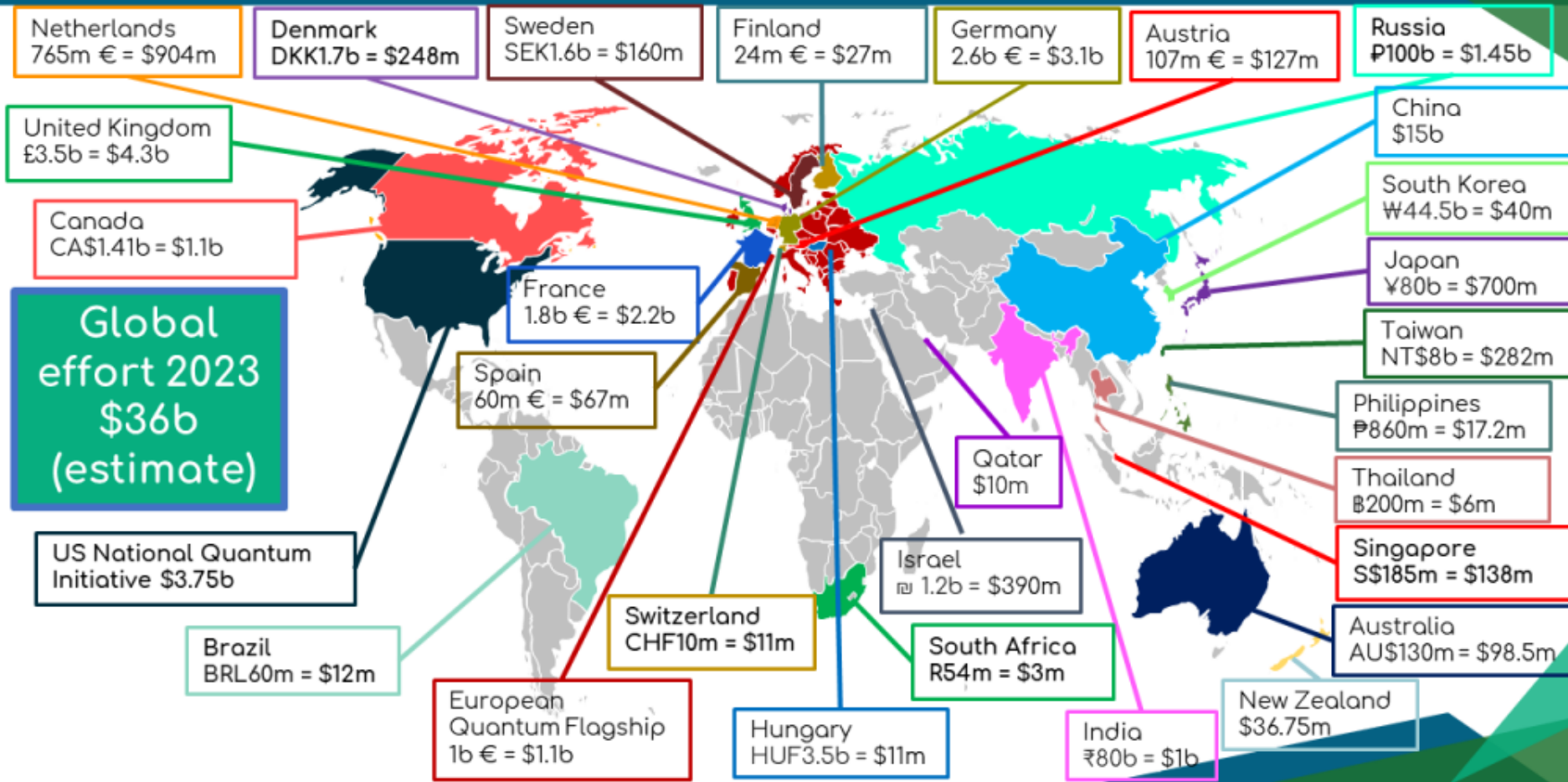
¹Total historic announced funding; timelines for investment of funding vary by country.
Source: Johnny Kung and Muriam Fancy, *A quantum revolution: Report on global policies for quantum technology*, CIFAR, April 2021; McKinsey analysis

McKinsey
& Company

To read the article, see ["Quantum computing funding remains strong, but talent gap raises concern," June 15, 2022.](#)

Investment in Quantum Computing Initiatives

Quantum effort worldwide



Source: <https://qureca.com/quantum-initiatives-worldwide-update-2023/>

International Quantum Conference hosted at WSU

- **1st Annual Quantum Technology for Defense conference**
- **8 and 19 May 2024**
- **Hosted at Wright State University**
- **Annual event where government agencies, technology leaders, academia and industry can meet to discuss quantum information science applied to defense**



**“When you change the way you look
at things, the things you look at
change.”**

**Max Planck, Quantum theorist and
Nobel Prize winner**

